

Calculus I Lesson 22

1) Equation (implicit function) : $5x^2 + y^2 = 12$

a) $D_x(5x^2) = ?$

b) $D_x(y^2) = \underline{\quad ? \quad} \cdot \frac{dy}{dx}$

c) $D_x(12) = ?$

d) Write the Implicit Differential Equation: $D_x(5x^2) + D_x(y^2) = D_x(12)$

e) Solve the equation in part (d) for $y' = \frac{dy}{dx} = \underline{\quad ? \quad}$

2) Equation (implicit function) : $\sqrt{x} + \sqrt{y} = 8$

a) $D_x(x^{1/2}) = ?$

b) $D_x(y^{1/2}) = \underline{\quad? \quad} \cdot \frac{dy}{dx}$

c) $D_x(8) = ?$

d) Write the Implicit Differential Equation: $D_x(\sqrt{x}) + D_x(\sqrt{y}) = D_x(8)$

e) Solve the equation in part (d) for $y' = \frac{dy}{dx} = \underline{\quad? \quad}$

3) Equation (implicit function) : $2x^3 - xy + y^2 = 12$

a) $D_x(2x^3) = ?$

b) $D_x(-x \cdot y) = -x \cdot D_x(y) + y \cdot D_x(x) = -x \cdot \frac{dy}{dx} + y \cdot 1 = \underline{\hspace{2cm}} ?$

c) $D_x(y^2) = \underline{\hspace{2cm}} ? \cdot \frac{dy}{dx}$

d) $D_x(12) = ?$

e) Write the Implicit Differential Equation: $D_x(2x^3) + D_x(-xy) + D_x(y^2) = D_x(12)$

f) Solve the equation in part (e) for $y' = \frac{dy}{dx} = \underline{\hspace{2cm}} ?$

4) Equation (implicit function) : $4x^3 - 3x^2 \cdot y + 2x \cdot y^2 = 10$

a) $D_x(4x^3) = \underline{\hspace{2cm}}?$

b) $D_x(-3x^2 \cdot y) = -3x^2 \cdot D_x(y) + y \cdot D_x(-3x^2) = \underline{\hspace{2cm}}?$

c) $D_x(2x \cdot y^2) = 2x \cdot D_x(y^2) + y^2 \cdot D_x(2x) = \underline{\hspace{2cm}}?$

d) $D_x(10) = \underline{\hspace{2cm}}?$

e) Write the Implicit Differential Equation: $D_x(4x^3) + D_x(-3x^2 y) + D_x(2x \cdot y^2) = D_x(10)$

f) Solve the equation in part (e) for $y' = \frac{dy}{dx} = \underline{\hspace{2cm}}?$

5) Equation: $x \cdot y = 4$

Find an equation of the tangent line passing through the point $(1,4)$.

a) $D_x(x \cdot y) = x \cdot D_x(y) + y \cdot D_x(x) = \underline{\hspace{2cm}}?$

b) $D_x(4) = \underline{\hspace{2cm}}?$

c) Write the Implicit Differential Equation: $D_x(x \cdot y) = 4$

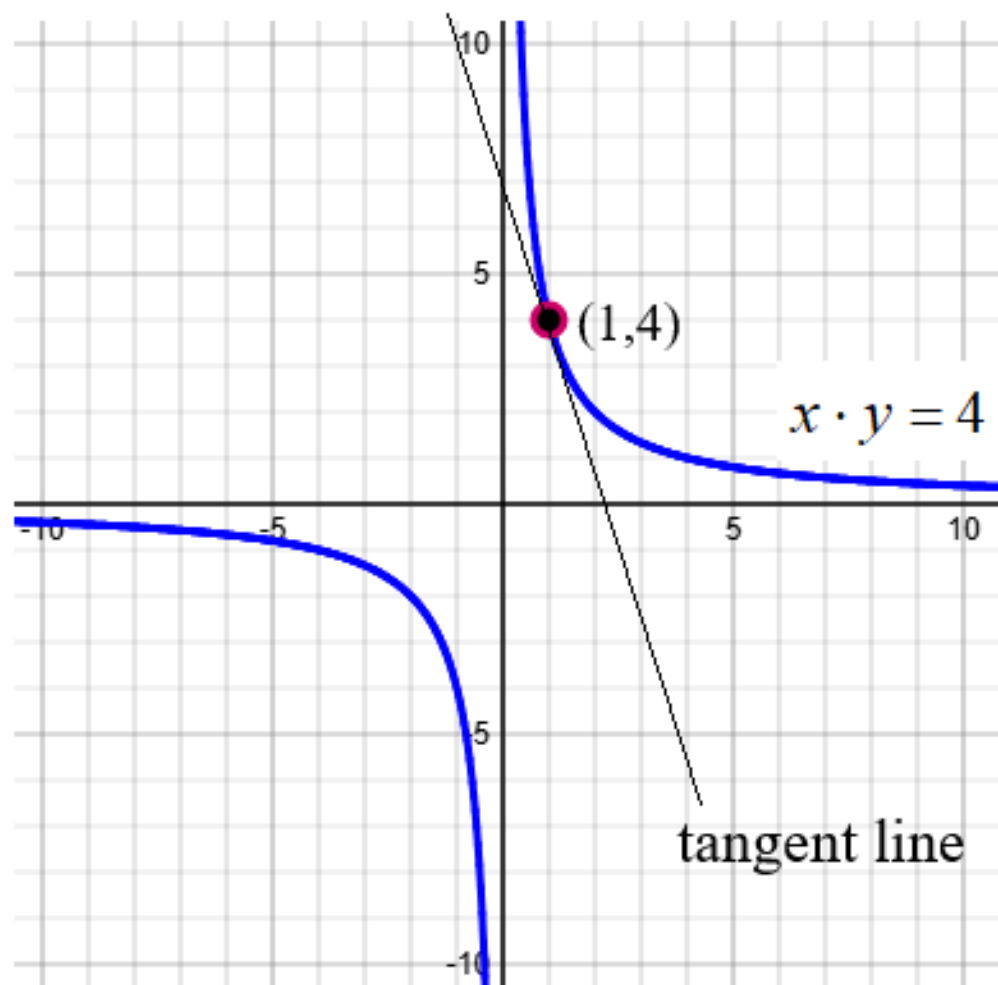
d) Solve the equation in part (c) for $y' = \frac{dy}{dx} = \underline{\hspace{2cm}}?$

e) Find the slope of the tangent line = $\underline{\hspace{2cm}}?$

Hint: Use the equation in part (e); replace x with 1 and y with 4.

f) Equation of tangent line is :

Hint: $y - y_1 = m(x - x_1)$.



6) Equation: $(3x^2 + 2) \cdot y = 8$

Find an equation of the tangent line passing through the point $(1, 8/5)$.

a) $D_x[(3x^2 + 2) \cdot y] = (3x^2 + 2) \cdot D_x(y) + y \cdot D_x(3x^2 + 2) = \underline{\hspace{2cm}}?$

Hint: $D_x(y) = 1 \cdot \frac{dy}{dx} = y'$

b) $D_x(8) = \underline{\hspace{2cm}}?$

c) Write the Implicit Differential Equation: $D_x[(3x^2 + 2) \cdot y] = D_x(8)$

d) Solve the equation in part (c) for $y' = \frac{dy}{dx} = \underline{\hspace{2cm}}?$

e) Find the slope of the tangent line = $\underline{\hspace{2cm}}?$

Hint: Use the equation in part (e); replace x with 1 and y with $8/5$.

f) Equation of tangent line is : $\underline{\hspace{2cm}}?$

Hint: $y - y_1 = m(x - x_1)$.

